

OS Threads & Synchronization Reference Sheet

Mechanical exam-oriented recap for mutexes, condition variables, semaphores, barriers, rwlocks and joins.

If you read...	Use
shared variable / shared array	mutex
alternate / turns / wait your turn	condition variable
all threads wait together	barrier
maximum N threads inside	semaphore
many readers, few writers	rwlock

1. MUTEX

Purpose:
Only ONE thread at a time.

Main functions:
`pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;`
`pthread_mutex_lock(&mutex);`
`/* critical section */`
`pthread_mutex_unlock(&mutex);`

Mental model:
Door with one key.

2. CONDITION VARIABLE

Purpose:
Put a thread to sleep until another thread wakes it.

Main functions:
`pthread_cond_wait(&cond, &mutex);`
`pthread_cond_signal(&cond);`
`pthread_cond_broadcast(&cond);`

Golden template:

```
pthread_mutex_lock(&mutex);
while(condition_false) {
    pthread_cond_wait(&cond, &mutex);
}
/* work */
pthread_cond_signal(&cond);
pthread_mutex_unlock(&mutex);
```

Mental model:
Bed + alarm clock.

3. SEMAPHORE

Purpose:
Allow exactly N threads simultaneously.

Main functions:
sem_t sem;

sem_init(&sem, 0, N);

sem_wait(&sem);

/* work */

sem_post(&sem);

Mental model:
Parking lot with N spaces.

4. BARRIER

Purpose:
All threads wait until everybody arrives.

Main functions:
pthread_barrier_init(&barrier, NULL, nr_threads);
pthread_barrier_wait(&barrier);

Mental model:
Race starting line.

5. RWLOCK

Purpose:
Many readers simultaneously, writers alone.

Reader:
pthread_rwlock_rdlock(&rwlock);

/* read */

pthread_rwlock_unlock(&rwlock);

Writer:
pthread_rwlock_wrlock(&rwlock);

/* write */

pthread_rwlock_unlock(&rwlock);

6. JOIN

Purpose:
Wait for another thread to finish.

Main function:
pthread_join(thread, NULL);

Difference:
join = one thread waits for another
barrier = everybody waits for everybody

7. Most Common Exam Patterns

Pattern A – mutex only

local computation
lock

```
update global  
unlock
```

Pattern B – condition variable

```
while(not my turn)  
    wait
```

work

```
change turn  
signal
```

Pattern C – barrier

phase 1

barrier wait

phase 2

Pattern D – semaphore

wait permit

work

return permit